

RESEARCH ARTICLE

Pore-Based Modeling of Hydraulic Conductivity Function of Unsaturated Rooted Soils

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ABSTRACT

Mualem's approach has been widely used to predict hydraulic conductivity functions (HCFs) of bare soils if a soil water retention curve (SWRC) model is available. The assumption that Mualem's approach holds is that the distribution of soil pores is spatially completely random. Under this assumption, relative hydraulic conductivity (K_r) is determined by the continuance probability of water-filled pores. However, this assumption is not valid for rooted soils, as root growth causes soil particle rearrangement, and thus soil pore rearrangement, altering the probability of pore connectivity. After reconsidering Mualem's assumption, this study attempts to develop a new approach for predicting HCF of rooted soils by modeling the root-induced pore rearrangement and the resultant change in the continuance probability of water-filled pores. Two approaches mentioned were incorporated with a root-dependent SWRC model to express HCF as a function of matric suction. The proposed model was validated against nine sets of measured HCFs from published studies. It was found that the proposed model reduced the root mean square error (RMSE) of K_r and $\lg K_r$ by 33% and 53%, respectively, as compared to traditional Mualem's model. Physically, the model's effectiveness depended on soil texture and root type. In fine-textured soils, roots were capable of displacing soil particles, thereby causing soil pore rearrangement. Also, coarse roots with high strength tend to alter pore distribution. After considering the effects of pore-level root-soil interaction on pore rearrangement, the proposed model provided a significant improvement in the prediction of HCF of unsaturated rooted soils.

1 | Introduction

Vegetated soil infrastructures, such as vegetated landfill covers, have gained increasing recognition in recent years in the discipline of eco-geotechnics [1, 2]. In vegetated soil infrastructures, soil behavior interacts closely with plant performances and atmo-

spheric conditions [3]. For instance, suction response and slope stability were influenced by the coupled effect of the temperature and roots, based on field monitoring and numerical models [4]. By altering the soil water conditions, rainwater infiltration influences the leachate and stability of the landfill body [5–7]. On one hand, rainfall permeates into the landfill sites and accelerates

the decomposition of the underlying municipal solid waste (MSW) [8]. This produces more contaminating leachate, which likely exceeds the capacity of purification facilities and thus poses a threat to groundwater [7]. After the selection of suitable plants, the rainwater percolation into the waste layer is minimized by (1) the transpiration-induced increase in soil suction and (2) root-induced mitigation in soil desiccation cracks [9]. On the other hand, the increase in soil water content leads to the large deformation and the reductions in soil suction and shear strength, which might ultimately cause catastrophic landslides [10]. Vegetation provides both mechanical and hydrological reinforcement for mitigating landslides induced by rainfalls [11]. In addition, vegetated soil layers are reported to reduce greenhouse gas emissions [12]. These functions mentioned are governed by the root-induced changes in soil hydraulic properties, especially the soil water retention curve (SWRC) and hydraulic conductivity function (HCF). Effects of root on soil microstructure alteration (e.g., pore clogging, generation of macro-pores, and amalgamation of micro-aggregates) and hence the soil hydraulic properties have been thoroughly reviewed by Lu et al. [13], based on published experiment data. However, there is still a research gap in modeling the HCF of unsaturated rooted soils with clear physical meanings.

For bare soils without vegetation, Mualem's approach has been the most widely used method to predict HCFs up to now [14, 15]. Mualem's approach is expressed as an integral formula with the term $d\theta/\psi$ (θ and ψ represent soil volumetric water content (VWC) and matric suction, respectively); thus, an SWRC (relationship of θ with ψ) model is needed to derive the closed-form analytical expression of K_r , for example, the van Genuchten SWRC model (VG SWRC) [16]. The predictions by the van Genuchten-Mualem (VG-M) model have been widely reported in studies on the bare soils without vegetation [16–18]. In the cases of rooted soils, the l -modified Mualem model has been used [19, 20]. The parameter l , which was fixed as 0.5 in the traditional Mualem's equation, was set as a material fitting parameter with a common range from -1.5 to 2.5 [19–21] in the l -modified Mualem model, in order to have a closer match with the measurements. The l -modified Mualem model improves the agreement between the measurements and the modeling by mathematically adding the degree of freedom, but it lacks physical nature.

To develop an HCF model of unsaturated rooted soils with clear physical meaning, the assumption of Mualem's approach might be reconsidered. The assumption that the Mualem's approach holds is that the spatial distribution of the soil pores is completely random. Based on this assumption, the probability of water-filled pore connectivity is calculated. However, the roots might alter soil pore connectivity through the creation, exploitation, and occupation of pore spaces [13]. Soil particle realignment induced by root growth causes the rearrangement of soil pores. The root-induced changes in pore spatial distribution have been widely observed in previous studies [22–25]. Roots with high biomass density caused the nonrandomness (i.e., heterogeneity) of the soil pore spacing [22]. Observations via x-ray computed tomography (CT), have shown that roots create new pores by pushing soil particles away [24]. Recently, the pore rearrangement induced by the roots has been observed by three-dimensional (3-D) CT imaging [25]. In addition to root-induced pore rearrangement, the occupancy of roots in pore space should be considered. For instance, Ng et al. [26] proposed a root-dependent SWRC model

that accounts for the changes in void ratio due to the root occupancy in soil pore space.

The root-induced pore rearrangement makes Mualem's approach invalid for rooted soils. Considering this invalidity, there remains a research gap in determining the HCFs of unsaturated rooted soils. Therefore, this study aims to propose a new model to predict HCFs of rooted soils, by considering the effects of not only root occupancy in pores but also root-induced pore rearrangement. The proposed model is validated against nine sets of data from the published studies and also compared with the traditional Mualem's approach.

2 | Model Formulation

In general, soils have three phases, namely soil particles (S), pore water (W), and pore air (A) (W and A belong to pore [P]), resulting in nine kinds of phase connections. Denote the connection of any two phases as connection $i-i$ ($i = S, W, \text{ and } A$). Among them, only pore water connection (i.e., W-W) permits water flow and thus controls the soil's hydraulic conductivity. Considering the probability of W-W connection, relative hydraulic conductivity (K_r ; Equation 1) may be represented by:

$$K_r = \frac{K_u}{K_s} \quad (1)$$

where K_u and K_s are the unsaturated hydraulic conductivity and saturated value, respectively.

In this study, Mualem's approach (Equation 2) [14] is used as a reference for developing the model for rooted soils. Considering spatially random distribution of soil pore distribution in Mualem's approach:

$$K_{r,M}(S_e) = S_e^l \left(\frac{\int_0^{S_e} dS_e/\psi}{\int_0^1 dS_e/\psi} \right)^2 \quad (2)$$

where $K_{r,M}$ is K_r following the Mualem's approach and is a function of the effective degree of saturation (S_e ; Equation 3); $l = 0.5$, as being initially suggested by Mualem [14] and commonly recommended by others [16, 17]:

$$S_e = \frac{\theta - \theta_{res}}{\theta_{sat} - \theta_{res}} \quad (3)$$

where θ is the soil's VWC; θ_{sat} and θ_{res} are the saturated and residual VWC, respectively.

To solve Equation (2), an expression relating S_e to matric suction ψ (i.e., SWRC) is needed. For bare soils, VG SWRC model [16] is adopted, as follows:

$$S_e = \left[1 + \left(\frac{\psi}{\alpha} \right)^n \right]^{-m} \quad (4)$$

where α is a parameter related to the soil's air entry value (AEV; unit: kPa); m and n are the dimensionless fitting coefficients, and generally assume $m = 1-1/n$.

For rooted soils, the root-dependent SWRC model (RO-SWRC model for short), previously proposed by Ng et al. [26], which considers root occupation in soil pore space, is adopted, as follows:

$$S_e = \left[1 + \left(\frac{e^{1/n}}{\alpha} \psi \right)^n \right]^{-m} \quad (5)$$

where e is the void ratio of the rooted soil, which can be expressed by Equation (6):

$$e = \frac{e_0 - R_v(1 + e_0)}{1 + R_v(1 + e_0)} \quad (6)$$

where e_0 is the void ratio of the bare soil, R_v is the root volume ratio (unit: cm^3/cm^3), which is defined as the ratio of root volume to the total soil volume. When $R_v = 0$, that is, $e = e_0$, the equation represents the SWRC of bare soil.

The combined use of Mualem's equation (Equation 2) with the VG SWRC equation (Equation 4) and with the RO-SWRC equation (Equation 5) is denoted as the VG-M and RO-M models, respectively. Substituting Equation (5) into Equation (2), the RO-M model can be expressed as follows (see detailed derivation in Appendix A):

$$K_{r,RO-M}(\psi) = \left\{ 1 + \left[\frac{e^{1/(n-1)}}{\alpha} \psi \right]^n \right\}^{\frac{1-n}{2n}} \times \left\{ 1 - \left\{ 1 - \left\{ 1 + \left[\frac{e^{1/(n-1)}}{\alpha} \psi \right]^n \right\}^{-1} \right\}^{1-1/n} \right\}^2 \quad (7)$$

Similarly, the VG-M model can be expressed as follows:

$$K_{r,VG-M}(\psi) = \left[1 + \left(\frac{\psi}{\alpha} \right)^n \right]^{\frac{1-n}{2n}} \times \left\{ 1 - \left\{ 1 - \left[1 + \left(\frac{\psi}{\alpha} \right)^n \right]^{-1} \right\}^{1-1/n} \right\}^2 \quad (8)$$

Any soil pore rearrangement induced by root growth is not considered in Mualem's approach. The growth of a root in the soil pore space may be represented by a simplified process as depicted in Figure 1. Considering two disconnected pores (named r and ρ) that are randomly distributed, the process of root growth can be divided into three stages. In stage I, the root tip enters and gradually fills the pore r (Figure 1a). In stage II, the root penetration upon growth would rearrange the soil particles and hence the pore space (Figure 1b); this process gradually reduces the size of pore ρ to nil and simultaneously creates a new pore ρ_2 continuous to the pore r (Figure 1c). In addition to soil particle rearrangement, the root-induced connected pores

could be due to cracking of macro-aggregates and amalgamation of micro-aggregates. Dynamic tensile stresses induced by root water uptake might cause macro-aggregates to break down into micro-aggregates, generating connected pores near these micro-aggregates [13]. Also, when the roots amalgamated the micro-aggregates, the micro-aggregates moved to each other, which rearranged pore distribution and caused connected pores around the amalgamation [13].

In this study, the HCF model is further modified to take into account any change in pore spatial distribution due to such a process of root penetration. Similar to the traditional Mualem's approach, the proposed approach makes four assumptions: (i) water enters soil from the smaller pores to the larger ones; (ii) the ratio of total pore area in any soil's cross-section to the section area is equal to soil porosity (n_{po}); (iii) soil pore is filled with either water or air, but not their mixture; and (iv) water is conductive only in W-W connection.

The soil pore that has a pore radius of r is denoted as pore r . Following assumption (i), a monotonic relationship between θ and r exists. For a given $\theta(r)$, soil pores with radii ranging from the minimal radius (R_{min}) to r are filled with water. The derivative of θ with respect to r is defined as the pore water distribution function $f(r)$, as follows:

$$f(r) = \frac{d\theta}{dr} \quad (9)$$

Integrating $f(r)$ for the entire range of r would return $\theta(r)$, as follows:

$$\int_{R_{min}}^r f(r) dr = \theta(r) \quad (10)$$

For a special condition of full saturation, soil pores with radii from R_{min} to R_{max} would be filled with water, as follows:

$$\int_{R_{min}}^{R_{max}} f(r) dr = \theta_{sat} = n_{po} \quad (11)$$

where R_{max} is the maximal pore radius.

Consider two adjacent thin soil layers, namely slabs X and Y . Following assumption (ii), the ratio of total pore area in any soil's cross-section to the section area is identical; thus, the porosity of slab X is equal to that of slab Y . Following the assumption made by Mualem [14] about the spatially random distribution of soil pore, the probability ($p_{P-P,bare}$) of P-P connection between slabs X and Y , in the case of bare soils, can be expressed as:

$$p_{P-P,bare} = n_{po}^2 = \theta_{sat}^2 \quad (12)$$

Substituting Equation (11) into Equation (12) returns the following equation:

$$p_{P-P,bare} = \int_{R_{min}}^{R_{max}} \int_{R_{min}}^{R_{max}} f(r) f(\rho) dr d\rho \quad (13)$$

where r and ρ is the pore radius in slab X and Y , respectively.

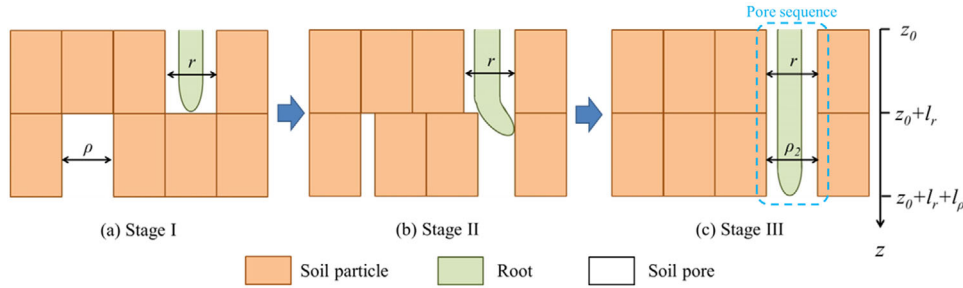


FIGURE 1 | Schematic illustration of the process by which the single root causes soil pore rearrangement.

At water content $\theta(R)$, the soil pore with radii less than R is filled with water following assumption (iii). In this case, the probability ($p_{W-W,bare}$) for water-filled pores with W-W connection in bare soils at $\theta(R)$ can be expressed as follows:

$$p_{W-W,bare}(R) = \int_{R_{min}}^R \int_{R_{min}}^R f(r)f(\rho)drd\rho \quad (14)$$

From previous experimental observation [22–25], physically, root growth could rearrange soil particles and hence pores, affecting the pore connectivity. Adopting the same form as Equation (14), the probability function ($p_{W-W,rooted}$) for water-filled pores with W-W connection in rooted soils at $\theta(R)$ can be expressed as:

$$p_{W-W,rooted}(R) = \int_{R_{min}}^R f(r)dr \quad (15)$$

where $p_{W-W,rooted}$ is the probability of W-W connection for the rooted soil in the proposed model.

This probability function attains its maximum value when the rooted soil is fully saturated (i.e., $\theta(R) = \theta(R_{max}) = \theta_{sat}$), as follows:

$$p_{W-W,rooted,max} = p_{W-W,rooted}(R_{max}) = \int_{R_{min}}^{R_{max}} f(r)dr \quad (16)$$

The relative ratio of pore-water-continuance (C_r) is defined as follows:

$$C_r(R) = \frac{p_{W-W,rooted}(R)}{p_{W-W,rooted,max}} = \frac{\int_{R_{min}}^R f(r)dr}{\int_{R_{min}}^{R_{max}} f(r)dr} \quad (17)$$

Following assumption (iv), water is conductive only in W-W connection. C_r is a measure of the pore water connectivity with respect to all the connected pores in a rooted soil. As suggested by Mualem [14], a parameter $T(R, r, \rho)$, which is a correction factor for soil tortuosity, is introduced to capture any change in soil tortuosity induced by the process of desaturation; another parameter $G(R, r, \rho)$, is introduced to correct the partial correlation between the pores r and ρ at $\theta(R)$. Accordingly, C_r becomes $C_{r,im}$:

$$C_{r,im} = \frac{\int_{R_{min}}^R T(R, r, \rho)G(R, r, \rho)f(r)dr}{\int_{R_{min}}^{R_{max}} T(R, r, \rho)G(R, r, \rho)f(r)dr} \quad (18)$$

where $C_{r,im}$ is the intermediate variable during the procedure of deriving K_r from C_r .

Following Mualem's approach [14], the product of r and ρ is introduced to consider the effects of actual flow configuration on K_r (see detailed derivation in Appendix B). Hence, Equation (18) becomes:

$$K_r = \frac{\int_{R_{min}}^R T(R, r, \rho)G(R, r, \rho)r\rho f(r)dr}{\int_{R_{min}}^{R_{max}} T(R, r, \rho)G(R, r, \rho)r\rho f(r)dr} \quad (19)$$

As a first approximation, it is assumed that root growth only affects the spatial distribution of pores, but not the pore size. Considering the relatively low values (about 1%) of the root volume ratio for all tested cases, pore size change induced by root occupancy may be limited. Mathematically, this assumption makes ρ constant and thus independent of dr . Hence, Equation (B5) becomes:

$$K_r = \frac{\int_{R_{min}}^R T(R, r, \rho)G(R, r, \rho)r f(r)dr}{\int_{R_{min}}^{R_{max}} T(R, r, \rho)G(R, r, \rho)r f(r)dr} \quad (20)$$

Due to the lack of any effective method to determine $T(R, r, \rho)$ and $G(R, r, \rho)$, these two functions were assumed to follow a power law with respect to θ or S_e , as reported in the literature [14, 27, 28]. Subsequently, Equation (20) becomes:

$$K_r = S_e^l \frac{\int_{R_{min}}^R r f(r)dr}{\int_{R_{min}}^{R_{max}} r f(r)dr} \quad (21)$$

where $l = 0.5$ for the case of bare soils [14, 27, 28]. Also, the product of G and T is assumed to follow a power function of S_e with $l = 0.5$ for the case of rooted soils [29, 30].

Recalling the Young–Laplace equation, pore radius r (unit: m) could be expressed as follows:

$$r = \frac{2\sigma \cos \beta}{\psi} \times 10^{-6} \quad (22)$$

where σ is the surface tension of water; β is the contact angle at the interface of fluid and solid ($^\circ$). Applying Equations (9) and (22), Equation (21) becomes:

$$K_r(\theta) = S_e^{1/2} \frac{\int_0^\theta d\theta/\psi}{\int_0^{\theta_{sat}} d\theta/\psi} \quad (23)$$

where $K_r(\theta)$ is the relative hydraulic conductivity of rooted soils at $\theta(R)$ following the proposed approach.

TABLE 1 | The comparison among three HCF models.

Model	SWRC equation	HCF equation	For bare or rooted soil	Consider root occupancy	Consider pore rearrangement
VG-M	van Genuchten [16]	Mualem [14]	Bare	No	No
RO-M	Ng et al. [26]	Mualem [14]	Rooted	Yes	No
RO-P	Ng et al. [26]	This study	Rooted	Yes	Yes

Abbreviations: HCF, hydraulic conductivity functions; SWRC, soil water retention curve.

Expressing VWC by S_e , Equation (23) becomes:

$$K_r(S_e) = S_e^{1/2} \frac{\int_0^{S_e} dS_e / \psi}{\int_0^1 dS_e / \psi} \quad (24)$$

Equation (24) can be solved by substituting the root-dependent SWRC model (Equation 5) and defining a dummy variable $y = S_e^{n/(n-1)}$:

$$\begin{aligned} K_r(S_e) &= S_e^{1/2} \frac{\int_0^{S_e} \left(\frac{S_e^{n/(n-1)}}{1 - S_e^{n/(n-1)}} \right)^{1/n} dS_e}{\int_0^1 \left(\frac{S_e^{n/(n-1)}}{1 - S_e^{n/(n-1)}} \right)^{1/n} dS_e} \\ &= S_e^{1/2} \frac{\int_0^{S_e^{n/(n-1)}} \left(\frac{y}{1-y} \right)^{1/n} \frac{n-1}{n} y^{-\frac{1}{n}} dy}{\int_0^1 \left(\frac{y}{1-y} \right)^{1/n} \frac{n-1}{n} y^{-\frac{1}{n}} dy} \\ &= S_e^{1/2} \left[1 - \left(1 - S_e^{n/(n-1)} \right)^{1-1/n} \right] \end{aligned} \quad (25)$$

Substituting Equation (5) into Equation (25), finally K_r expressed as a function of ψ can be given as follows:

$$\begin{aligned} K_{r,RO-P}(\psi) &= \left\{ 1 + \left[\frac{e^{1/(n-1)}}{\alpha} \psi \right]^n \right\}^{\frac{1-n}{2n}} \\ &\times \left\{ 1 - \left\{ 1 - \left\{ 1 + \left[\frac{e^{1/(n-1)}}{\alpha} \psi \right]^n \right\}^{-1} \right\}^{1-1/n} \right\} \end{aligned} \quad (26)$$

where $K_{r,RO-P}$ denotes the K_r of the rooted soils using the root-dependent SWRC model via the proposed pore-based model, named RO-Pore (RO-P) model.

In addition to the effect of root occupancy in pore space on HCF, as considered by the RO-M model, the proposed RO-P model further accounts for the root-induced change in pore arrangement. To predict the HCF of bare soils, the parameter e_0 should be measured, while the parameters α and n should be calibrated against any available SWRCs. For the case of rooted soils, an additional parameter, R_r , is required for the prediction of HCF. Table 1 summarized the differences among the three HCF models mentioned above.

3 | Database and Statistics

Twelve sets of measured data of HCFs were collected from four published studies [30–33] to evaluate the accuracy of HCF prediction (as shown in Table 2). The database included nine cases of rooted soil (S1–S9) and three cases of bare soil (B1–B3). B1, B2, and B3 were the bare soil cases corresponding to S1, S2–S4, and S8–S9, respectively, whereas no bare soil data was available for S5–S7. Some basic physical properties of the test soils, including their classification following ASTM [34], are summarized in Table 3.

All the selected cases were the direct laboratory measurements of HCF by the instantaneous profile method [35]. The VG model was used to fit the SWRCs of the bare soils, and then their HCFs were predicted by the VG-M model. The parameters required for solving the VG's SWRC and VG-M HCF models are summarized in Table 4.

For the cases of rooted soils, the root-dependent SWRC model was used to fit the SWRCs reported in the respective literature, and then the HCF of the rooted soils was predicted by both the RO-M and RO-P models. For a fair comparison of each case, the identical sets of parameters were used in these two models. The parameters used for the root-dependent SWRC, RO-M HCF, and RO-P HCF models are summarized in Table 5. Note that for S5, S8, and S9, both α and n values were obtained from the respective literature. For cases S1–S4, α was also adopted from the source study, while n was determined by best-fitting. Because the SWRCs of the bare and rooted soils were not available in S6 and S7, their parameters, α and n , were obtained from the back-calculation of the HCF of the respective bare soils. The best-fitted values of α and n were obtained after fitting the measured HCF data using the VG-M model. As a result, these two parameters could under- or over-estimate the hydraulic properties of the rooted soils. The K_r might be underestimated due to the underestimation of α for a given suction, according to Equation (26).

The root mean square error (RMSE) was used as a statistical measure to evaluate the model's predictability. Two kinds of RMSEs, that is, RMSE (K_r) and RMSE ($\lg K_r$), were calculated via Equations (27) and (28), respectively. During the calculation of RMSE (K_r), K_r at low suction plays a predominant role, because it could be about two orders of magnitude greater than K_r at relatively high suction. Thus, RMSE ($\lg K_r$) is recommended to evaluate the effectiveness of model prediction throughout the entire measured suction range [37].

$$RMSE(K_r) = \sqrt{\frac{\sum_{i=1}^{i=N} (K_{r,m,i} - K_{r,p,i})^2}{N}} \quad (27)$$

TABLE 2 | Database used for model validation.

No.	Soil classification	Vegetation	Specimen size	Reference
S1	Low-plasticity silt	Vetiver (<i>Chrysopogon zizanioides</i>)	100 mm diameter; 150 mm height	[31]
B1	Low-plasticity silt	No	100 mm diameter; 150 mm height	
S2	Clayey sand	Vetiver	100 mm diameter; 150 mm height	
S3	Clayey sand	Vetiver	100 mm diameter; 150 mm height	
S4	Clayey sand	Vetiver	100 mm diameter; 150 mm height	
B2	Clayey sand	No	100 mm diameter; 150 mm height	[30]
S5	Silty gravel with sand	Vetiver	200 mm diameter, 200 mm height	
S6	Sandy lean clay	Vetiver	240 mm length, 240 mm width, and 200 mm height	[32]
S7	Sandy lean clay	Bermuda (<i>Cynodon dactylon</i>)	240 mm length, 240 mm width, and 200 mm height	[32]
S8	Sandy lean clay	Wheat (<i>Triticum aestivum</i> L.)	65 mm diameter, 95 mm height	[33]
S9	Sandy lean clay	Okra (<i>Abelmoschus esculentus</i> (L.) Moench)	65 mm diameter, 95 mm height	
B3	Sandy lean clay	No	65 mm diameter, 95 mm height	

TABLE 3 | Some basic physical properties of the tested soils in the selected database.

No.	Gravel (%)	Sand (%)	Silt (%)	Clay (%)	Dry density (g/cm ³)	Liquid limit (%)	Plastic limit (%)	Plasticity index
S1, B1	0	20.2	51.8	28	1.34	39	29	10
S2–S4, B2	31.4	45.7	15.4	7.5	1.43	31	19	12
S5	31.3	28.8	26.9	13	N/A	N/A	N/A	N/A
S6, S7	0	30	56	14	1.38	35	16	19
S8, S9, B3	0	48	44	8	1.30	42	20	22

TABLE 4 | Parameters used in VG-M model for bare soils.

No.	K_s (m/s) ^a	α (kPa) ^b	n^c	$m = 1-1/n$
B1	5.11×10^{-8}	2.3	1.090	0.083
B2	2.90×10^{-7}	0.7	1.118	0.106
B3	1.80×10^{-9}	15.6	1.130	0.115

Abbreviation: VG-M, van Genuchten-Mualem.

^aMeasured maximum hydraulic conductivity ($K_{\max}$) based on the instantaneous profile method.

^bObtained from the respective literature.

^cObtained through best-fitting.

where $RMSE(K_r)$ is the RMSE value of K_r ; N is the number of measured data points and i is a counter; $K_{r,m,i}$ and $K_{r,p,i}$ are measured and predicted K_r , respectively.

$$RMSE(\lg K_r) = \sqrt{\frac{\sum_{i=1}^{i=N} [(\lg K_{r,m,i}) - (\lg K_{r,p,i})]^2}{N}} \quad (28)$$

where $\lg K_r$ is the logarithmic function of K_r with the base of 10; $RMSE(\lg K_r)$ is the RMSE of $\lg K_r$.

4 | Results and Discussion

Before validating the proposed model, (a) the measured SWRCs of the bare soils (B1–B3) were fitted by the VG's SWRC model; (b) HCFs of the bare soils (B1–B3) were predicted by the VG-M model; and (c) the SWRCs of the rooted soils (S1–S9) were fitted by the root-dependent SWRC model. As the proposed RO-P model is developed on the basis of the VG-M model, it is crucial to first show the capability of this model to capture the SWRC and the predictability on HCF of bare soils from (a) and (b). Also, the RO-P model of HCF is derived from the RO-SWRC model [26], so the fitting of the SWRC by this model needs to be ensured, from (c). Thus, through the procedures (a), (b), and (c), the RO-P model can be more fairly evaluated. Finally, statistically quantitative comparisons between the RO-P and RO-M models were conducted.

4.1 | Fitted SWRCs and Predicted HCFs of Bare Soils

Figure 2 depicts the comparison between the measured and fitted SWRCs of the selected bare soil cases. As expected, the VG's SWRC model can capture the measured data well, especially for

TABLE 5 | Parameters used in RO-M and RO-P models for rooted soils.

No.	K_s (m/s)	α (kPa)	n	e_0 (cm ³ /cm ³)	R_v (cm ³ /cm ³)
S1	^a 3.46×10^{-7}	^a 1.8	^b 1.236	^a 1.020	^c 1.60×10^{-2}
S2	^a 5.80×10^{-6}	^a 0.9	^b 1.598	^a 0.911	^c 1.17×10^{-2}
S3	^a 3.68×10^{-6}	^a 0.9	^b 1.621	^a 0.911	^c 1.10×10^{-2}
S4	^a 4.02×10^{-7}	^a 0.8	^b 1.683	^a 0.911	^c 1.14×10^{-2}
S5	^a 7.80×10^{-8}	^a 19.8	^a 1.466	^a 0.832	^a 1.85×10^{-3}
S6	^d 5.04×10^{-7}	^e 8.35	^e 7.970	^a 1.000	^a 7.81×10^{-3}
S7	^d 1.00×10^{-7}	^e 8.35	^e 7.970	^a 1.000	^a 2.41×10^{-3}
S8	^d 6.23×10^{-9}	^b 9.76	^b 1.180	^a 0.750	^f 7.40×10^{-3}
S9	^d 3.40×10^{-9}	^b 14.48	^b 1.220	^a 0.750	^f 3.16×10^{-3}

Note: The parameter m is calculated by $1-1/n$.

^aObtained from the respective literature.

^bObtained through best-fitting.

^cAssuming the specific root dry biomass is equal to 0.5 g/cm³ [36].

^dMeasured maximum hydraulic conductivity (K_{max}) based on the instantaneous profile method.

^eVia the back-calculation of the measured HCF of bare soils.

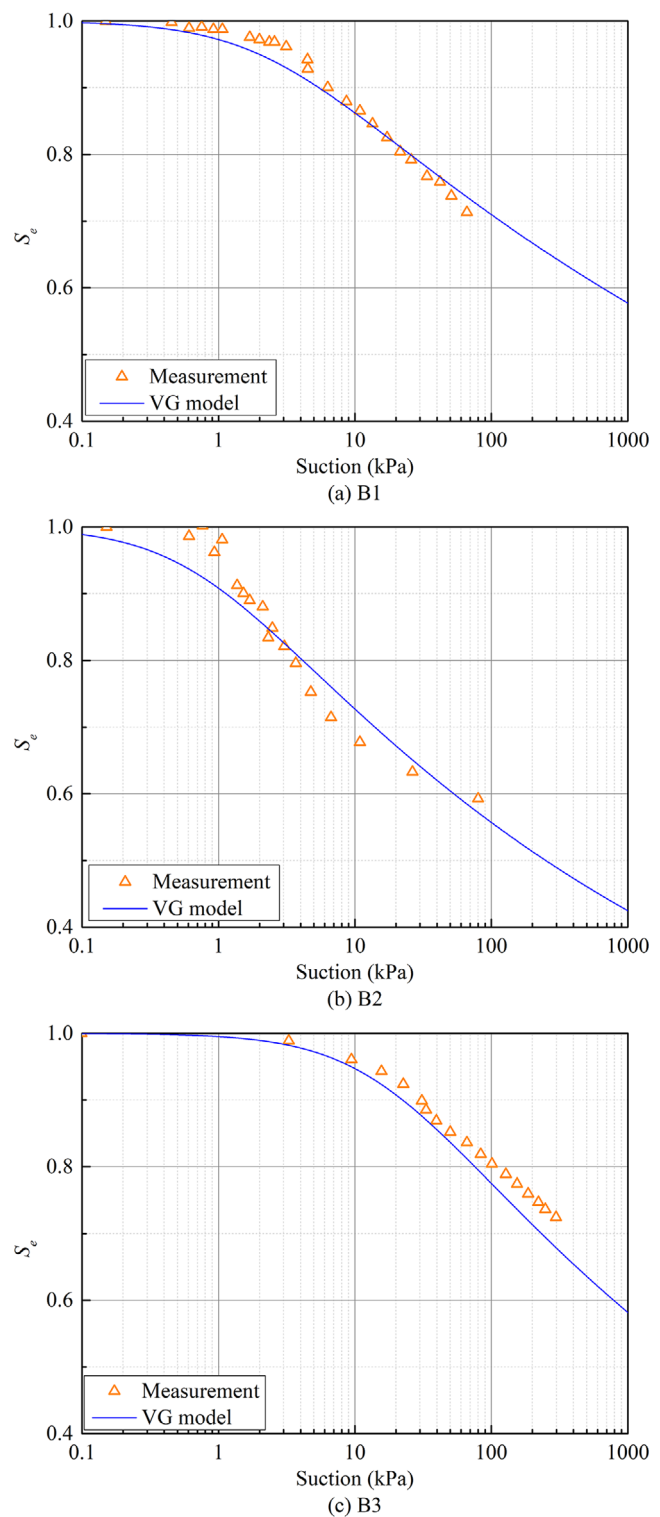
^fBased on the measured root diameter distribution and assuming all root sections are circular.

case B1 (low-plasticity silt). For case B2 (clayey sand; corresponding rooted soil cases: S2–S4), the VG model underestimated S_e in the near-saturation suction range (< 3 kPa), but overestimated S_e in the medium suction range (near 10 kPa). In addition, the VG model slightly underestimated S_e in the high suction range (> 100 kPa) for case B3 (sandy lean clay; corresponding rooted soil cases: S8, S9). In general, the VG model can describe the SWRCs of bare soil well and thus represents a relevant model for the prediction of the SWRC of rooted soils using the RO-SWRC model.

Figure 3 depicts the comparison between the measured and predicted HCFs of the bare soil cases based on the VG-M model using the fitting parameters of the VG's SWRC model. For all cases, the VG-M model only qualitatively captured the trend (i.e., the lower K_r at higher suction), but failed to quantitatively predict the magnitude of the HCFs. Rather large discrepancies existed between the measured and predicted HCFs for all three cases. The discrepancy might be because the bare soils in these selected cases have significant pore-level heterogeneity associated with the compaction, soil suffusion, and irrigation, and so forth, which would lead to nonrandom pore distribution [38–42], violating the spatially random soil pore distribution assumption made by the Mualem's approach.

4.2 | Fitted SWRCs of Rooted Soils by the Root-Dependent Model

Because both the RO-M model and the proposed RO-P model used the root-dependent SWRC model (RO-SWRC for short) to predict HCF, it was essential to study the performance of the RO-SWRC model before evaluating the accuracy of the prediction of the HCF made by the two models. To capture the effects of root

**FIGURE 2** | Fitting of the measured SWRCs of the selected bare soil cases by the VG's SWRC model. SWRC, soil water retention curve; VG, van Genuchten.

occupation of soil pores, the RO-SWRC model was used to fit the reported data from seven selected cases of rooted soils (except S6 and S7, when such data were unavailable).

Figure 4 shows the comparison of the measured and fitted SWRCs of the selected rooted soil cases. Evidently, RO-SWRC model can

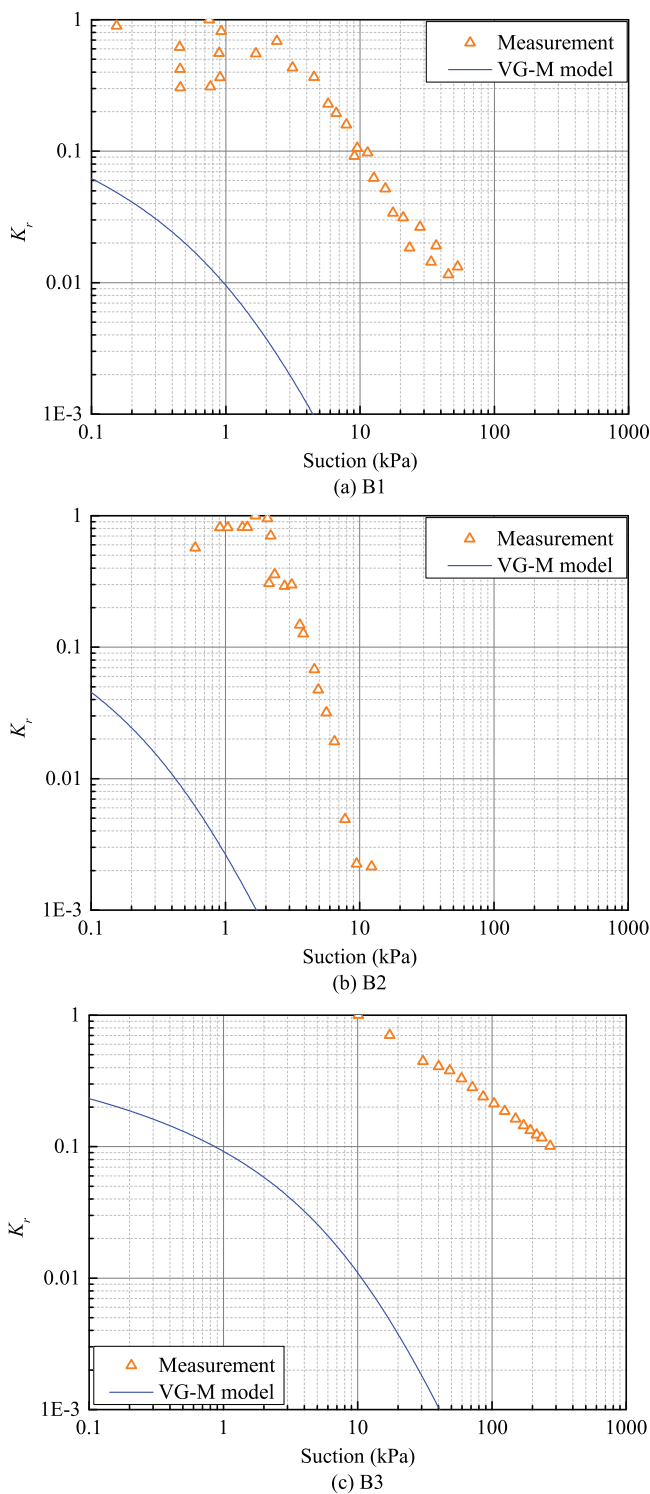


FIGURE 3 | Prediction of the HCFs of the selected bare soil cases by the VG-M model. VG-M, van Genuchten-Mualem.

fit the measured data of SWRCs well in most cases, except (1) near-saturation situations of S2–S4 and (2) S8 as well as S9. The model underestimated the AEV for cases S2–S4. The reductions in measured S_e were much more significant than in the fitted values near a suction of 1 kPa. All three cases considered clayed sand vegetated with Vetiver grass, which has coarse roots. Coarse roots altered the distribution of pore size by generating macropores [13]. After the generation of macro-pores, pores with larger

sizes could dehydrate at lower suction, that is, air could more easily enter the soil pores, according to Young–Laplace equation. This likely caused the observed sharp reduction in measured S_e , which was not considered in the root-dependent SWRC model. All evaluations highlighted that the root-dependent SWRC model of rooted soil provided a relevant basis to develop the RO-M and RO-P HCF models.

4.3 | Validation of Newly Proposed Model

The accuracy of the prediction of HCF made by the RO-M model and the proposed RO-P models was assessed by comparing the predictions with the corresponding measurements, as depicted in Figures 5 and 6 for cases of fine-textured soils and coarse-textured soils, respectively. Generally, the measured HCFs of rooted soils spanned more orders of magnitude than the corresponding bare soils. For instance, the HCF for case B1 spanned two orders of magnitude, while that for case S1 spanned three orders of magnitude. Further, as both the RO-M and RO-P models shared identical sets of parameters, the only determining factor to affect model effectiveness was the difference between Equations (A4) and (26), that is, physically, whether the effects of root-induced pore rearrangement were considered or not. In general, both models qualitatively captured the trend (i.e., lower K_r at higher suctions). The proposed RO-P model appeared to make a better prediction than the RO-M model, probably because of the consideration of the additional mechanism of root-induced pore rearrangement (Figure 1).

For case S1, the RO-P model performed better than the RO-M model. However, the RO-P model did not capture the shape of the measured HCF well. This discrepancy could be because of the limited suction range considered in the experiments [31], and the lack of information about residual suction and residual water content for soils with relatively high clay content could be a reason for causing the mismatch with the HCF measurements. The prediction improvement was much more significant for the cases S1 and S5–S9. In these five cases out of six, fine-grained soils (i.e., fine grain fraction > 50%) were considered. Low-plasticity silt and sandy lean clay were tested in S1 and S6–S9, respectively. Even if coarse-textured soil was used in S5, it contained about 40% fines. The dependency on soil texture was probably because roots more easily alter soil pore distribution within fine-textured soils, as compared to coarse-textured soils. In coarse-grained soils, roots tended to clog soil pores, instead of rearranging them [13, 43]. Roots were expected to buckle under surrounding stress from coarse particles due to the lack of strength to displace them [44, 45]. In this case, root penetration upon growth was mainly along the tortuous pathways formed by the coarse particles and finally clogged the soil pore, which might reduce saturated hydraulic conductivity [13, 23, 46]. Effects of the pore clogging mechanism on the saturated hydraulic conductivity could not be captured by the proposed HCF model. In fine-grained soils, on the contrary, roots were capable of rearranging surrounding soil particles during the growth process [13, 47] and altered the pore connectivity [22–25]. Therefore, the proposed model displayed a better improvement of the prediction of the HCF through the additional consideration of the mechanisms of root-induced pore rearrangement, especially for soils with a high content of fines.

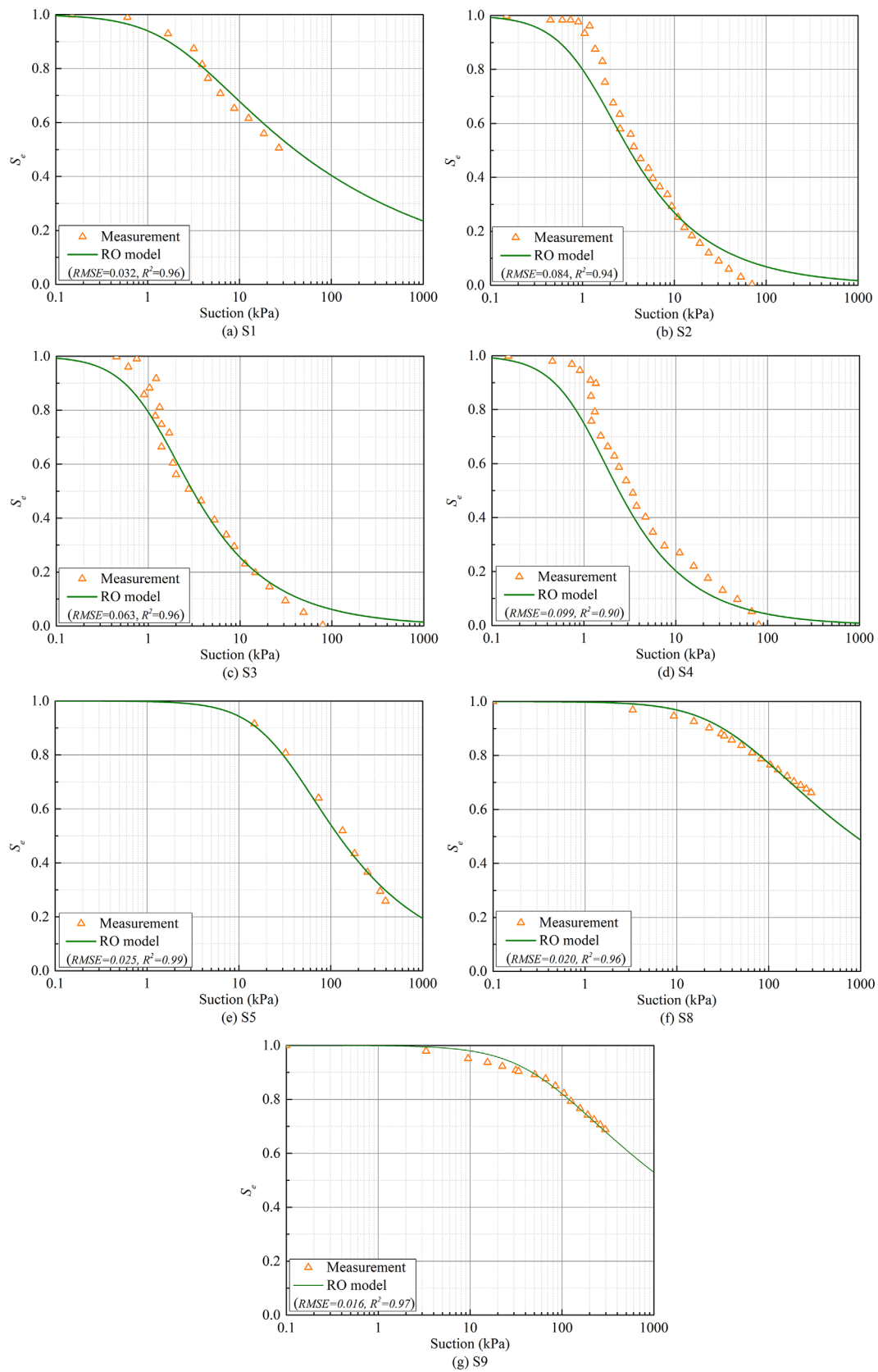


FIGURE 4 | Fitting of root-dependent SWRC model that considers root occupancy of soil pore space (cases S6 and S7 are not available). SWRC, soil water retention curve.

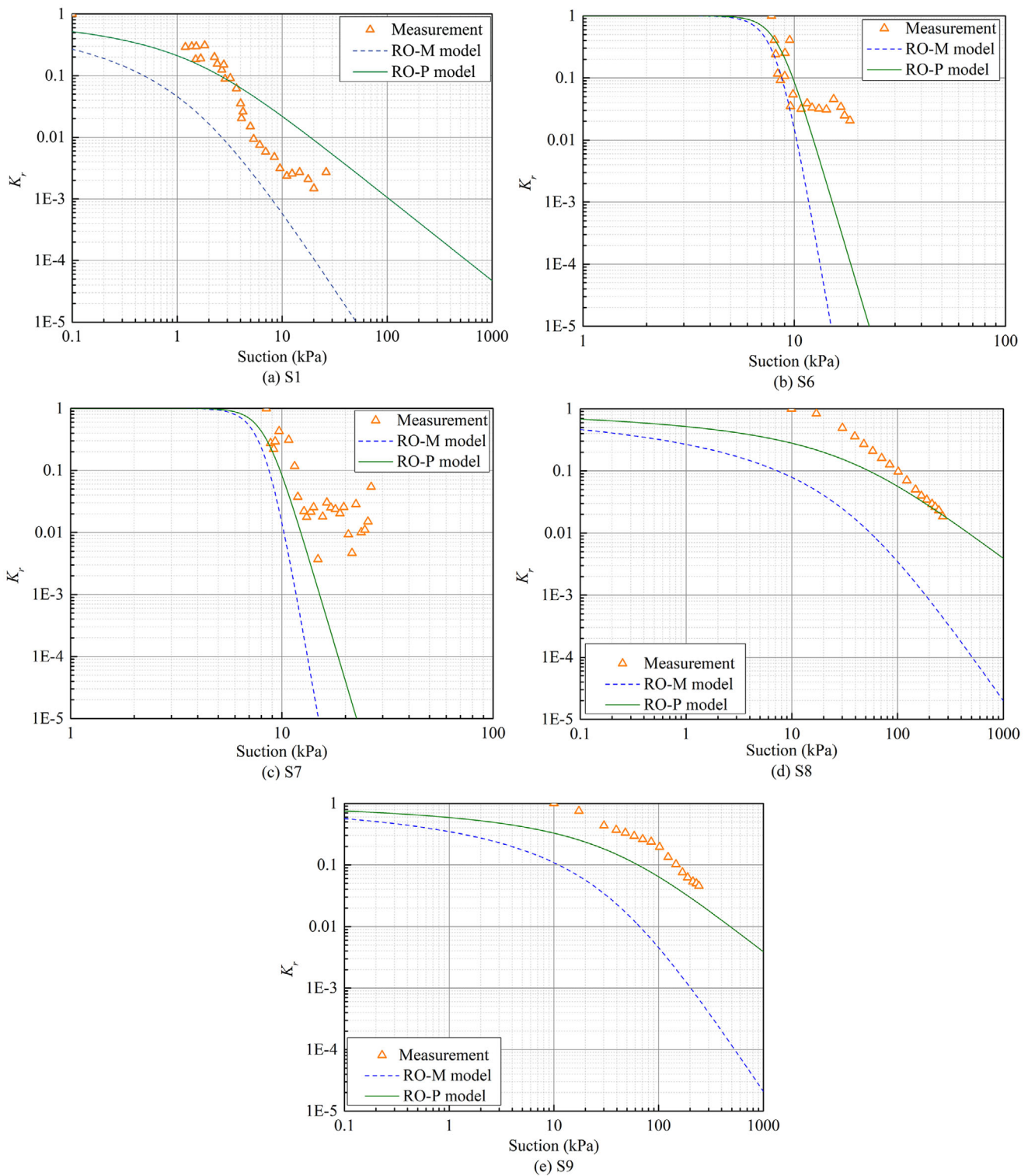


FIGURE 5 | The comparisons among the measurements and the predictions by RO-P model and RO-M model for fine-textured soils.

4.4 | Assessment of the Predictability of the Proposed RO-P Model

The values of RMSE (K_r) and RMSE ($\lg K_r$) and their averages (Ave.) and standard deviations (SD) are given in Figures 7 and 8, respectively. From Figure 7, an overall comparison of RMSE (K_r) under all nine cases suggested that the RO-P model (Ave. = 0.145; SD = 0.064) could better capture the HCFs of unsaturated

rooted soils, as compared to the RO-M model (Ave. = 0.193; SD = 0.096). The newly proposed model reduced the RMSE (K_r) by 33%. As shown in Figure 7, the RO-M model performed slightly better than the RO-P model in the coarse-textured soil cases S3 and S4. For coarse-textured soils except S3 and S4, however, the RO-P model could predict better than the RO-M model. The failure of the RO-M model was because nonrandomness might be induced by many processes (e.g., compaction and irrigation) in

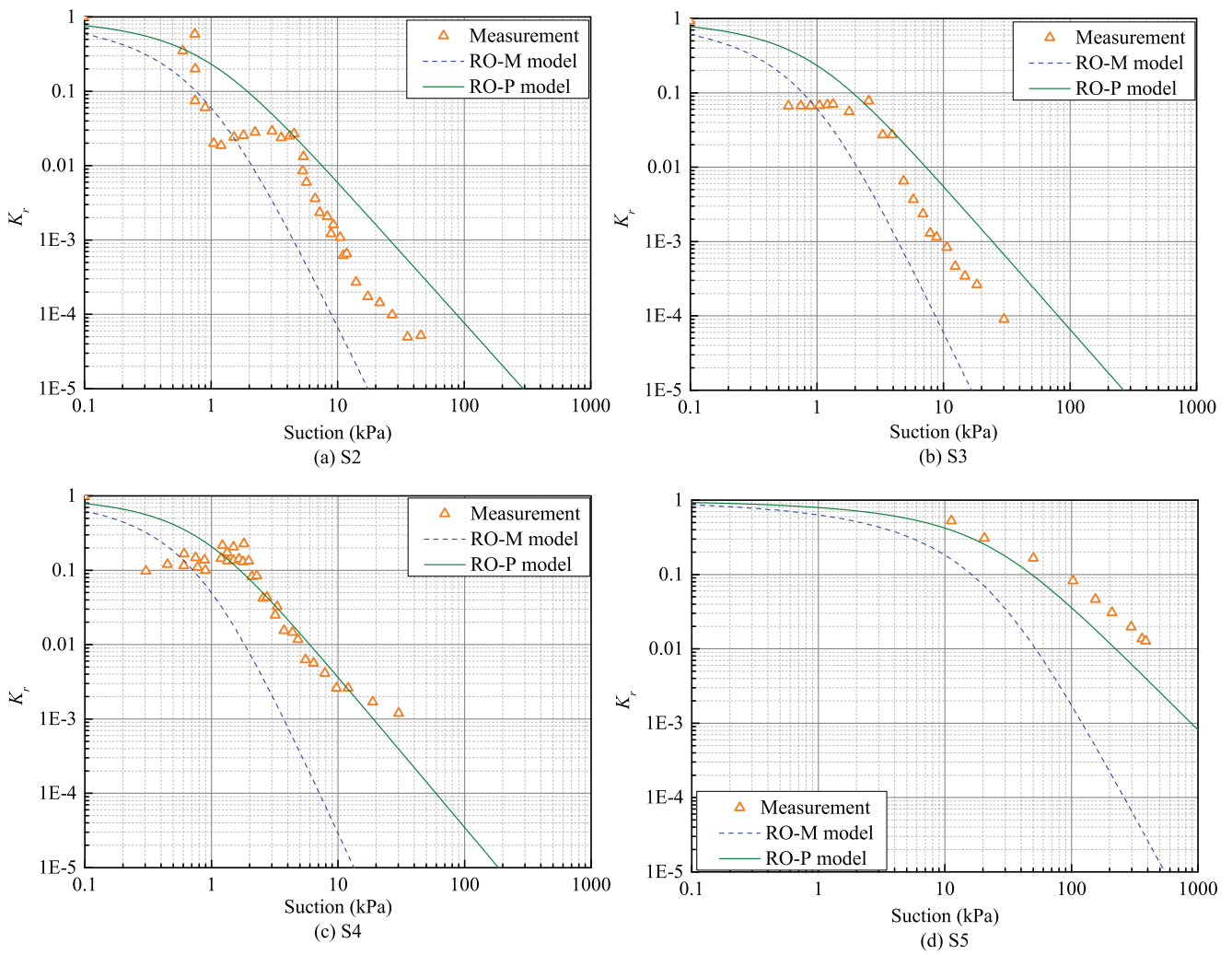


FIGURE 6 | The comparisons among the measurements and the predictions by RO-P model and RO-M model for coarse-textured soils.

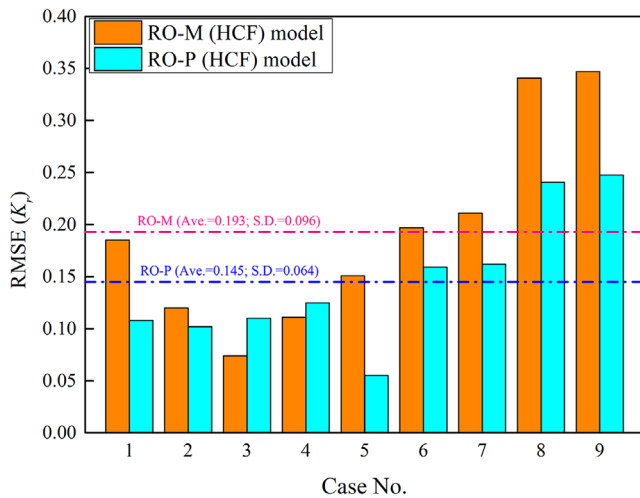


FIGURE 7 | RMSE (K_r) values and their averages and standard deviations of two models. RMSE, root mean square error.

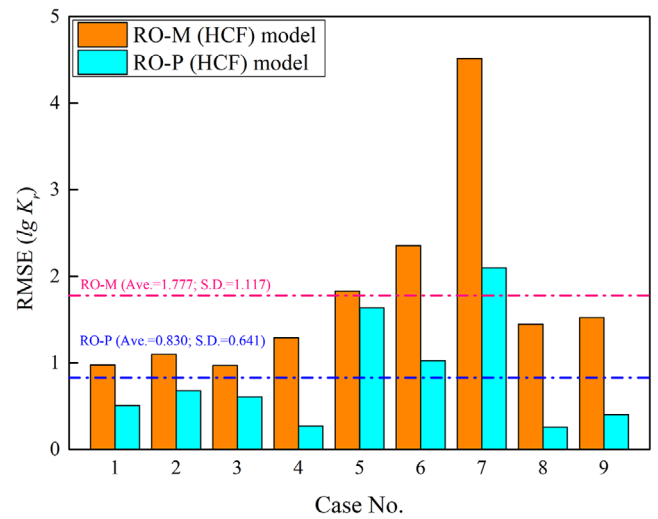


FIGURE 8 | RMSE ($\lg K_r$) values and their averages and standard deviations of two models. RMSE, root mean square error.

addition to the root growth [38–40]. As presented in Figure 8, the predictions made by the RO-M model were far from the respective measurements. Seven of the nine cases showed the RMSE ($\lg K_r$)

> 1 for the prediction by the RO-M model. Compared to the RO-M model, the RO-P model generally reduced the RMSE ($\lg K_r$) by

53%. The failure of Mualem's approach was not surprising due to the lack of consideration of root-induced pore rearrangement.

Physically, the difference between the predictabilities of RO-P and RO-M models was influenced by the soil texture and the root type. In fine-textured soils, root growth tends to rearrange the fine grains in the vicinity, which might create more continuous pores along the roots [13, 24, 47]. In coarse-textured soils, on the contrary, root growth tends to leave soil particles in their original locations due to the lack of capacity to displace the coarse grains [13, 43, 45]. In addition, root type was found to play a secondary role in affecting model effectiveness. Based on the RMSE ($\lg K_r$), the RO-P model performed more satisfactorily in cases S1-S4 (Vetiver-vegetated cases), S8 (Wheat-vegetated case), and S9 (Okra-vegetated case) than the others. The highest RMSE ($\lg K_r$) was observed for case S7 when considering Bermuda grass. In general, the roots of Bermuda are finer than other roots, especially the roots of Vetiver [32]. Coarser roots have a greater ability to displace the soil particles, leading to local compression as well as realignment of soil particles and further soil pore rearrangement [13]. Furthermore, the lowest RMSE (K_r) was observed in the case S5. However, the RMSE ($\lg K_r$) of case S5 (i.e., 1.633) was much higher than the average of the nine cases (i.e., 0.830). This observation suggested that the RO-P model can predict hydraulic conductivity at near-saturation conditions well, but fails to capture the hydraulic conductivity at higher suctions in this case. This phenomenon suggested that the model prediction depended on the suction in case S5 (silty gravel with sand). In this case, the abilities of roots to rearrange coarse particles were limited throughout the entire suction range. At near-saturation (i.e., low suction), roots might have broken some aggregates formed by the fine particles during root growth (i.e., penetration and thickening), causing rearrangement of fine particles and soil pores. However, this soil-root interaction was relatively more difficult at high suction.

In the proposed pore-based model, all soil pores were assumed to be continuous between adjacent soil layers because of root growth. In reality, however, only a portion of the pores (instead of all of them) may be rearranged by the roots. Thus, a combination of the proposed model and Mualem's model was expected to perform better than the proposed model or Mualem's model. Given this, it was required to first determine how many pores were influenced by roots. This might be a function of root volume ratio, root diameter distribution, and root morphological traits, and so forth; however, relevant data were not available. The HCF prediction of rooted soils could be further improved if the relevant root data are made available or reported in future studies.

5 | Conclusions

This study proposed a new approach to predict HCF of the rooted soils. Mualem's and the new approaches were incorporated into a root-dependent SWRC model and then compared against nine sets of measured HCF data from the published studies. The following conclusions may be drawn:

1. Only pore water connection permits the water flow and therefore controls HCF of the soils. The root growth could rearrange the soil particle and thus pore distribution, altering

the continuance probability of water-filled pores. Given this, the physical nature of the proposed model is to consider the effects of not only root occupancy in soil (or pore-clogging mechanism) but also pore rearrangement induced by root growth on HCF by calculating the continuance probability of water-filled pores before and after such pore structure changes.

2. The newly proposed model has shown the ability to effectively predict HCFs of unsaturated rooted soils. As compared to traditional Mualem's model, the proposed model reduced the RMSE (K_r) and RMSE ($\lg K_r$) by 33% and 53%, respectively, when the model took into account the effects of roots on the soil pore distribution and the following continuance probability of water-filled pores. A reduction of RMSE ($\lg K_r$) up to 53% has shown the significant improvement in HCF prediction of rooted soils by the proposed model throughout the entire suction range, after considering the mechanism of root-induced pore rearrangement.
3. The prediction on HCF was affected by the fitting of the corresponding SWRC model against the measured SWRC. Physically, the effectiveness of HCF prediction was influenced by the soil texture and the root type. In fine-textured soils, roots were capable of displacing soil particles and thus caused soil pore rearrangement, altering the probability of pore connectivity. More satisfactory model effectiveness was observed in cases of roots with high-density coarse roots with enough strength to alter soil particle and pore distributions.

Author Contributions

Hao Wang: conceptualization, methodology, visualization, formal analysis, writing—original draft. **Rui Chen:** funding acquisition, supervision, validation, writing—review and editing. **Anthony Kwan Leung:** funding acquisition, supervision, writing—review and editing. **Ankit Garg:** validation, visualization, writing—review and editing. **Zhenliang Jiang:** validation; writing—review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Based on Equation (5), $1/\psi$ was expressed as a function of S_e :

$$\frac{1}{\psi} = \frac{e^{1/(n-1)}}{\alpha} \left(\frac{S_e^{\frac{n}{n-1}}}{1 - S_e^{\frac{n}{n-1}}} \right)^{1/n} \quad (\text{A1})$$

Substituting Equation (A1) into Equation (2) returns:

$$K_{r,M}(S_e) = S_e^{1/2} \left[\frac{\int_0^{S_e} \left(\frac{S_e^{\frac{n}{n-1}}}{1 - S_e^{\frac{n}{n-1}}} \right)^{1/n} dS_e}{\int_0^1 \left(\frac{S_e^{\frac{n}{n-1}}}{1 - S_e^{\frac{n}{n-1}}} \right)^{1/n} dS_e} \right]^2 \quad (\text{A2})$$

Define a dummy variable $y = S_e^{n/(n-1)}$, Equation (A2) can be resolved as follows:

$$\begin{aligned} K_{r,M}(S_e) &= S_e^{1/2} \left[\frac{\int_0^{S_e^{\frac{n}{n-1}}} \left(\frac{y}{1-y} \right)^{1/n} \frac{n-1}{n} y^{-\frac{1}{n}} dy}{\int_0^1 \left(\frac{y}{1-y} \right)^{1/n} \frac{n-1}{n} y^{-\frac{1}{n}} dy} \right]^2 \\ &= S_e^{1/2} \left[1 - (1 - S_e^{\frac{n}{n-1}})^{1-1/n} \right]^2 \end{aligned} \quad (\text{A3})$$

Substituting Equation (5) into Equation (A3) returns Equation (A4):

$$\begin{aligned} K_{r,RO-M}(\psi) &= \left\{ 1 + \left[\frac{e^{1/(n-1)}}{\alpha} \psi \right]^n \right\}^{\frac{1-n}{2n}} \\ &\times \left\{ 1 - \left\{ 1 - \left\{ 1 + \left[\frac{e^{1/(n-1)}}{\alpha} \psi \right]^n \right\}^{-1} \right\}^{1-1/n} \right\}^2 \end{aligned} \quad (\text{A4})$$

Define the pore sequence as the group of two connected pores r and ρ_2 in the slabs X and Y (as shown in Figure 1c). Recalling the Poiseuille's law (Equation B1) and Darcy's law (Equation B2).

$$\Delta P = \frac{8\mu LQ}{\pi R_e^4} \quad (\text{B1})$$

where ΔP is the pressure difference between two ends of the pore sequence; μ is the dynamic viscosity; L is the length of the pore sequence; Q is the volumetric flow rate; R_e is the equivalent radius of the pore sequence and its value will be discussed below:

$$K = \frac{QL}{\pi R_e^2 \Delta h} = \frac{QL\rho_w g}{\pi R_e^2 \Delta P} \quad (\text{B2})$$

where K is the hydraulic conductivity; Δh is the hydraulic head difference; ρ_w is the water density; g is the gravitational acceleration.

Substituting Equation (B1) into Equation (B3) returns:

$$R_e^2 = \frac{8\mu K}{\rho_w g} = \frac{8\nu K}{g} \quad (\text{B3})$$

where ν is the kinematic viscosity (i.e., the ratio of dynamic viscosity to water density).

Parameter g is a constant, so it can be seen that K is proportional to R_e^2 at a given ν . Although ν is a variable, it plays no role in affecting K_r , because its effects on K_s and K_u are of the same magnitude. R_e^2 has been expressed in many studies. Mualem's expression (1976) was used in this study, as follows:

$$R_e^2 = r\rho \quad (\text{B4})$$

Add R_e^2 into the expressions of K_u and K_s , Equation (18) becomes:

$$K_r = \frac{\int_{R_{\min}}^R T(R, r, \rho) G(R, r, \rho) r \rho f(r) dr}{\int_{R_{\min}}^{R_{\max}} T(R, r, \rho) G(R, r, \rho) r \rho f(r) dr} \quad (\text{B5})$$